XuanEr (Seona) Rao

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RESEARCH INTERESTS

Human–Computer Interaction; Computational Design; Interactive Media;Gamification;Inclusive design;Digital Fabrication; AIGC

EDUCATION

National University of Singapore, Singapore **Aug 2026-Jul 2027 (Expected)**
Master of Science in Engineering Design & Innovation

Macau University of Science and Technology, Macau **Sep 2022-Jun 2026**
Bachelor of Fine Arts in Digital Media

- **Graduated with First-Class Honors:** Awarded the highest honors with a Scholarship by Melco Resorts (Macau) limited to 2023, 2024, and 2025 respectively, Dean's Honors List (3 times)

PUBLICATIONS

Gamification and cognitive factors: Research hotspots, knowledge structure, and future directions based on bibliometric analysis. Journal of Intelligence
Song, D., Guo, J., ***Rao, XE***., Hu, X., Gu, X., & Chen, J. (2026). *Journal of Intelligence, 14(7), 150.*
[SSCI,JCR Q1] *https://doi.org/10.3390/jintelligence14070150*

RESEARCH EXPERIENCE

Immersive Reality Lab, National University of Singapore
Research Assistant | Aug 2026–Present | Supervised by Dr Shaoyu Cai

- **VacGuide (leading; ongoing):** Investigating how passive internal guide geometry programs angular motion through vacuum-driven membrane collapse; developed 5-mm open-guide and multi-guide primitives spanning 0–180° rotations within a shared pneumatic pouch, and is characterizing geometry–motion relationships toward inverse design of articulated shape-changing structures.
- **Haptic Biofeedback Wearable For Therapistsupported Selfregulation (co-leading; ongoing):** Designing an EDA, PPG and IMU based wearable with vibrotactile and pneumatic feedback to support physiological self-regulation and therapist-mediated co-regulation during therapeutic conversations.

Digital Innovation and Humanities Education Research Center,Macau University of Science and Technology
Research Intern | Jun 2025–May 2026 | Supervised by Prof. Junming Chen

- Contributed to AIGC,Generative adversarial networks(GANs) for Bibliometric Review of AI-Generated Applications in Gamification Using CiteSpace.resulting work available as Journal of Intelligence, 14(7), 150. [SSCI,JCR Q1] https://doi.org/10.3390/jintelligence14070150

WORK EXPERIENCE

NetDragon Websoft Inc. – Experience Designer (AIGC Production) | **Jun-Sep 2024 Fujia,China**
Qingbo Intelligent Technology Co.Ltd. – Design Intern (UIUX Design)| **Apr-Jun 2024 Beijing,China**

SKILLS

- Prototyping & Fabrication: Pneumatic prototyping, 3D printing, heat sealing, soldering, electronics integration, rapid prototyping
- Hardware & Embedded: Arduino/ESP32, Lily Pad,E-textiles,sensors, circuit prototyping
- Programming & Design: Python, C/C++, LLM,MATLAB, Rhino, Unreal Engine 5,Unity, Figma,3DMax,Blender

AWARDS

- **2024 CMIT:** National Gold Award
- **16th User Experience Design Award** : First price of south china division（User experience Professional Association）
- **12th National College Digital Art&Design Awards:** Second price of Macau division
- **3rd Beijing Internet Audiovisual Arts Conference:** National third price

ACADEMIC REFEREES

Dr. Shaoyu Cai | Lecturer, NUS | shaoyucai@nus.edu.sg
Dr. Aleksandar Kostadinov | Lecturer, NUS | aleks_id@nus.edu.sg